

1st Youth International Math Olympiad

Division 2

Test A

Format: Questions are Integer Answers [0,999].

Scoring: Q1–5: 4 pts each Q6–15: 5 pts each Q16–20: 7 pts each

Time: 80 minutes

- Five squares are drawn with side lengths of 1, 2, 3, 4, and 5 units, respectively. What is the sum of their areas?
- Determine the value of the expression $(2 - 0 + 2 - 6)^{(2+0+2+6)}$.
- Towns A and B are 180 kilometers away from each other. Alice leaves Town A at 8:00 AM, traveling at 30 kilometers per hour. Betty leaves Town B some time after 8:00 AM, traveling at 60 kph. They meet each other at 11:00 AM. How many minutes after 8:00 AM did Betty leave Town B?
- It's 2019 and James Harden is averaging 40 points per game! Assuming he makes 12 free throws (1 point each) per game and makes twice as many 2-point shots as 3-pointers, how many 3-pointers does he average per game?
- Determine the smallest positive integer n such that all divisors of 2025 are not relatively prime to n .
- For the system of equations in x and y :

$$x + 2y = 15, \quad ax + y = 14.$$

If the solutions x and y are both natural numbers, find the sum of all possible values of the natural number a .

- Within the coordinate plane, a circle passes through the points $(0,0)$, $(6,0)$, and $(0,8)$. Find the radius of the circle.
- Six people sit at two distinguishable tables such that at least two people sit at each table and person A cannot sit at the same table as person B . How many ways are there to distribute the five people?
- The quadratic $x^2 - bx + 6 = 0$ has two positive roots c and d with $c - d = 5$. Find $c^2 - d^2$.
- Three squares with side lengths 3, 4, 5 are stacked vertically so that their left edges are aligned, with the 3×3 square on top, the 4×4 square in the middle, and the 5×5 square at the bottom. A straight line is drawn from the top left corner of the 3×3 square to the bottom right corner of the 5×5 square. The length of the portion of the line that lies inside the 4×4 square can be written as $\frac{m}{n}$ where m and n are coprime positive integers. Find $m + n$.

11. Suppose $P(x)$ is a monic quadratic polynomial with two different roots α and β , such that regardless of how the roots are labelled, we have $\alpha^2 + 6\beta = 29$. Find $P(0)$.

(Note that a *monic* quadratic polynomial is a quadratic with the coefficient of the highest degree term (x^2) equal to 1.)

12. Determine the unique value of a such that the roots of the polynomial $x^3 - ax + 120$ are all integers.
13. Consider rolling a fair die 5 times. The probability that the sum of the top faces is exactly 18 can be written as $\frac{m}{n}$, where m and n are relatively prime. What is the value of $m + n$?
14. Suppose that a and b are positive integers such that $\frac{20}{a} + \frac{26}{b} = 1$. Find the sum of all possible values of a .
15. The sum of the first 2026 terms of a geometric sequence is 162. The sum of the first 6078 terms is 342. Find the sum of the first 4052 terms.
16. When a certain natural number is prime-factorized, it becomes $2^5 \times 3^4 \times 5^3 \times 7^2$. Among the positive divisors of this number, find the sum of all divisors whose ones digit is 1.
17. Let k denote the minimum positive integer such that $n+3$ divides $n^2 + 6n + k$ for all $1 \leq n \leq 10$. Find the largest prime divisor of k .
18. The sequence $\{a_n\}$ is defined by $a_1 = 1$ and

$$a_{n+1} = \frac{a_n}{1 + n a_n} \quad \text{for all } n \geq 1.$$

Find the value of $\frac{1}{a_{45}}$.

19. Among quadratics $P(x)$ with positive leading coefficient c , $P(1) = 0$ and $P(P(2)) = 0$, the sum of the non-1 roots is 3. With $c = a + \sqrt{b}$, find $a + b$.
20. The value of

$$\prod_{j=10}^{35} (1 + \tan j^\circ)$$

can be expressed in the form 2^k . Find k .

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Test B

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Scoring: Q1–5: 4 pts each Q6–15: 5 pts each Q16–20: 7 pts each

Time: 80 minutes

1. In Project Sekai, the Append chart of MarbleBlue has 2500 notes and lasts 2 minutes 43 seconds. This means that to clear the chart, an average of k notes are pressed each second. Determine the largest integer at most k .
2. Shining writes the decimal representations of the powers of 2 in increasing order starting with 1. He continues until he writes a value containing d for some digit $0 \leq d \leq 9$. What choice of d allows him to write the maximum possible number of digits?
3. For some sequence of positive real numbers, the sum of the first n terms of a sequence is given by $3n^2 + 5n$. What is the 10th term of the sequence?
4. What is the area of a rectangle with diagonals of length 10 and a perimeter of 28?
5. Let x and y be independently and uniformly chosen real numbers from the interval $[-1, 1]$. What is the probability that $x^2 + y^2 \leq 1$?
6. Call a pair of integers (a, b) with $0 \leq a, b \leq 6$ *lucky* if the number $2^a \cdot 3^b$ is divisible by 12. Find the number of *lucky* pairs.
7. Billy randomly chooses an integer from the interval $(0, 2026)$. Bob randomly chooses an integer from the interval $(0, 4052)$. What is the probability that Billy's number is greater than Bob's?
8. The polynomial $x^3 - 6x^2 + 11x - 6 = 0$ has roots a, b, c . The value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be expressed as $\frac{m}{n}$ in lowest terms. Find $m + n$.
9. How many positive integers less than 1000 are divisible by both 7 and 11, but not divisible by 2?
10. Find the largest integer n such that 3^n is a divisor of $100!$.
11. Alice randomly chooses a value between 1 and 400 inclusive, and tells Bob the remainder when her value is divided by 2, 3, 5, and 7 respectively. Bob also knows that the value Alice chose is between 1 and 400 inclusive. The probability that Bob is able to uniquely determine the value Alice chose can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Determine $a + b$.
12. We define $f(n) = n - \lfloor \sqrt{n} \rfloor^2$ for positive integers n . Compute the sum $\sum_{n=1}^{100} f(n)$.
13. Find the sum of all positive integers n for which the value of $n^2 + 10n + 100$ is a perfect square.

14. How many six-digit positive integers have the property that the sum of their digits is equal to 45?
15. Find the 135th positive integer whose decimal representation contains only odd digits.
16. Let S be the set of all positive integers n such that the decimal representation of $n!$ ends in exactly 100 zeros. What is the sum of the elements in S ?
17. How many positive integers between 1 and 999 inclusive have digits that sum to less than their product?
18. The roots of $f(x) = x^2 - \frac{\sqrt{10}}{2}x + q$ are $\sin \theta$ and $\cos \theta$ for some real number θ . A second quadratic polynomial $g(x) = x^2 + mx + n$ has roots $\tan \theta$ and $\cot \theta$. The value of $|m \cdot n|$ can be expressed as $\frac{p}{q}$ in lowest terms. Find $p + q$.
19. The infinite series $S = \sum_{n=3}^{\infty} \frac{100}{n^2 - 4}$ converges to a rational number that can be expressed as an irreducible fraction $\frac{p}{q}$. Find $p + q$.
20. Suppose a basketball team has 12 players, 5 of which are on the court and the other 7 are available as substitutes. Define a substitution as replacing 1 player on the court with a player on the bench. Assume the coach can make a maximum of 5 substitutions per game and no player may re-enter after being substituted out. Let n be the number of ways the coach can make substitutions during the game if the coach does not need to use all 5 substitutions and the order of substitutions matter. Find $n \pmod{1000}$.

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Division 1

Test A

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Time: 80 minutes

1. Consider the right triangle ABC with $AB = 3$, $BC = 4$, and $AC = 5$. Define point D on line AB such that CD is the angle bisector of $\angle ACB$, and point E on line AB such that $\angle ACE = 3\angle ACD$. Find the distance from the incenter of triangle ABC to line AE .
2. A right circular cone has a base radius of 6 and a slant height of 10. A sphere is inscribed inside the cone. A second, smaller sphere is inscribed in the cone so that it rests directly on top of the first sphere and is internally tangent to the lateral surface of the cone. The volume of the second sphere can be strictly written as $\frac{m\pi}{n}$, where m and n are relatively prime positive integers. Find $m + n$.
3. Consider a 3×4 grid of squares where each unit square is colored black or white. In an operation, we select a 2×2 subgrid and flip the color of each of its squares. Define a coloring to be *magnificent* if it is possible to make all unit squares white after a series of operations. How many magnificent colorings are there?
4. The positive integer N is less than 1000 and contains no trailing zeroes. If the leftmost digit of $11N$ is 7, the second-leftmost digit of $29N$ is 8, the third-leftmost digit of $55N$ is 2, and the second-leftmost digit of $89N$ is 6, what is N ?
5. Let $S(n)$ denote the sum of the base-10 digits of the positive integer n . Find the remainder when the smallest positive integer n satisfying $S(n) + S(n+1) + S(n+2) = 207$ is divided by 1000.
6. Find the sum of all integers $n < 10,000$ such that the last four digits of n and n^2 are the same.
7. Triangle ABC has $AB = 13$, $BC = 14$, $AC = 15$. Find $[ICI_A]$, where I denotes the incenter and I_A is the excenter opposite A (the center of the excircle tangent to side BC and the extensions of sides AB and AC).
8. Bessie and Elsie are playing hide and seek on the coordinate plane. Bessie chooses a lattice point (m, n) with $0 \leq m, n \leq 60$ to hide, and barriers are placed at every other lattice point. Each second, Elsie moves one unit in some cardinal direction and removes the barrier at the point at which she arrives. After Elsie moves, if the line of sight between Elsie and Bessie doesn't contain any barriers, Elsie finds Bessie and Bessie earns a number of points equal to the number of seconds that have elapsed. Otherwise, the game continues. Let the point (a, b) be the point at which Bessie should hide to maximize the number of points she is guaranteed to get. Find $a + b$.
9. Let $0 \leq x \leq \frac{\pi}{2}$ be the solution to $\frac{\sin(2x) \cos(2x)}{\sin x \cos x} = \frac{1}{2026}$. What is the value of $\frac{\tan(2x)}{\tan x}$?

10. Two circles O_1 and O_2 have radii 5 and 12, respectively, and the distance between their centers is 13. The circles intersect at two points A and B . A line is drawn through point A , intersecting circle O_1 at C and circle O_2 at D , such that A lies strictly between C and D . What is the maximum possible area of triangle BCD ?
11. Consider a random string of length 2026 containing 0s and 1s. Find the expected length of the maximum subsequence of the binary string such that 010 and 101 are not contained within the subsequence.
12. There exists a unique pair of positive real values (a, b) with $a + b = 67$ such that $x^2 + ax + b$ has real roots and the minimum positive value c such that $x^2 + (a + c)x + b$ has real roots is equal to the maximum positive value d such that $x^2 + ax + (b + d)$ has real roots. The value of a can be expressed as $\alpha\sqrt{\beta} - \gamma$ for positive integers α, β, γ with β square-free. Find $\alpha + \beta + \gamma$.
13. A standard deck of 52 playing cards consists of 13 Spades, 13 Hearts, 13 Diamonds, and 13 Clubs. The deck is shuffled, and cards are drawn face up, one at a time, without replacement. Let E , expressed as a fraction in simplest form $\frac{m}{n}$, be the expected number of non-Spade cards drawn strictly before the 4th Spade appears. Find $m + n$.
14. Squares S_1 and S_2 are placed into the plane such that they share part of a side. The rectangle with the centers of S_1 and S_2 as opposite vertices is divided by the sides of S_1 and S_2 into four regions, three of which are squares and one of which is a rectangle R . It is known that the extension of one of the diagonals of R passes through a vertex of S_1 . The sum of the possible ratios of the area of S_1 to the area of S_2 can be expressed as $\frac{a+\sqrt{b}}{c}$ for positive integers a, b, c with b squarefree. Find $a + b + c$.
15. Consider the set of all quadratic polynomials $P(x)$ with positive leading coefficient c such that $P(1) = 0$ and $P(P(2)) = 0$. It is known that the sum of the possible values of the root of $P(x)$ not equal to 1 is 3. The value of c can be expressed as $a + \sqrt{b}$ for positive integers a and b . Find $a + b$.
16. Let N be the number of monic polynomials $P(x)$ of degree 10 with only real roots such that $P(x)P(-x) = P(x^2)$ for all real x . The sum of all possible values of $P(2)$ can be expressed as $\frac{m}{n}$ where m and n are relatively prime. Find $m + n$.
17. Let S be the set of all positive integers $n < 2187$ such that the representation of n in base 3 consists only of digits from the set $\{0, 1\}$. For each n in S , let $f(n)$ be the value of n when its base-3 digits are instead interpreted as a base-5 number. Find the number of elements n in S such that $f(n)$ is a multiple of 7.
18. A machine has 6 toggle switches, all initially in the “off” position. Every second, the machine selects one of the 6 switches uniformly at random and flips its position. The expected number of seconds until all 6 switches are simultaneously in the “on” position for the first time can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.
19. Let $P(x) = x^{20} + 5x^3 + 7x^2 + 1$ have roots $r_1, r_2, \dots, r_{20}$. The value of $\prod_{k=1}^{20} \frac{1}{r_k^4 + r_k^2 + 1}$ can be expressed as $\frac{m}{n}$ where m and n are relatively prime, find $m + n$.
20. Let ABC be a triangle with side lengths $AB = 13$, $BC = 14$, $CA = 15$. Let N be the Nagel point of ABC (the intersection of three lines connecting each vertex to the point of tangency of

the opposite excircle on the sides of the triangle). Let D, E, F be the midpoints of sides BC, CA , and AB , respectively. X, Y , and Z are midpoints of segments AN, BN , and CN . Find the area of hexagon $FXEZDY$.

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Division 1

Test B

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Time: 80 minutes

1. LeBron James needs to move 67 times to get from Ohio to Skibidi City. LeBron also needs to move $x + 67$ times to get from Skibidi City back to Ohio. LeBron is able move on average 6.7 times a day. If it takes LeBron 67 less days to get from Ohio to Skibidi City than back, determine $\lfloor x \rfloor$.
2. How many ways are there to arrange the letters in NEXTSTEM such that no two same letters touch each other?
3. A standard chess king is placed on a square of an infinitely large chessboard at $(0, 0)$. In a single valid move, the king can step to any of the 8 adjacent squares. Let N be the number of distinct paths consisting of exactly 9 valid king moves that start at $(0, 0)$ and end at $(7, 7)$. Find the remainder when N is divided by 1000.
4. Alice and Bob start on opposite vertices of a regular 6-gon. Each second, Alice moves along a random edge, and Bob moves along a random diagonal of the 6-gon that is not an edge. The expected number of seconds until Alice and Bob meet on a vertex at least once can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Find $a + b$.
5. Let r, s, t be the roots of $x^3 + 2x^2 + 4x + 9$. Determine the value of $r^4s^4 + s^4t^4 + r^4t^4$.
6. 9 cards with the values $1, 2, \dots, 9$ are placed on a table face down in some unknown order. Each second, a random card is turned over. If at any point the face-up cards contain some two coprime integers, the process terminates. The expected number of seconds it takes for the process to stop can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Find $a + b$.
7. Let ABC and DEF be two equilateral triangles with the same center such that DEF is contained inside ABC . Let I and J be the intersections of lines DE and DF with BC . If I and J are points on segment BC such that $BJ = 2$, $JI = 3$, and $IC = 1$, the length of EF can be expressed as $\frac{a\sqrt{b}}{c}$ where a, b, c are positive integers with a and c relatively prime and b squarefree. Find $a + b + c$.
8. Consider a dartboard of 5 unit squares formed by a unit square surrounded by 4 unit squares connected by edges. At the start of each second, if we have 3 darts on the board, we remove the dart that was thrown the earliest. Then, another dart is thrown at a random unit square which does not already have a dart. The expected number of seconds until darts are in 3 unit squares that are on a line can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Find $a + b$.

9. Let $ABCD$ be a square of side length 1. Let M and N be points on BC and CD respectively with $BM = ND$. Assume that P is a point on DN and I and J are the intersections of BP with AM and AN respectively. If $IJ = AJ$, determine the maximum possible measure of $\angle BAM$ in degrees.
10. In triangle ABC , the side lengths are $AB = 13$, $BC = 14$, and $CA = 15$. Let O be the circumcenter and H the orthocenter. The square of the distance OH^2 can be expressed as a fraction $\frac{m}{n}$ in simplest form. Find $m + n$.
11. There are 2026 balls in a bowl, each ball can be either red or blue. Welu chooses the number of balls that are red by selecting an integer at random from the set $\{0, 1, 2, \dots, 2026\}$. After Welu chooses the red balls you decide to draw 1000 balls at random without replacement. Given that all the balls you drew were red, the probability that all 2026 balls in the bowl were red is $\frac{N}{M}$ where N and M are coprime. Find $N + M$.
12. Consider a circle of radius 1 with center O . Let P and Q be points outside the circle such that if OP and OQ intersect the circle at I and J , we have that IQ and PJ are both tangent to the circle. If $3IJ = PQ$ and IQ intersects PJ at R , the value of OR^2 can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Find $a + b$.
13. Let n be the smallest positive integer such that the equation $x^2 + y^2 = n(x + y)$ has exactly 2025 positive integer solutions (x, y) . Find the number of positive integer divisors of n .
14. Jimmy Johnny has a fair seven-sided die with faces numbered 1, 2, 3, 4, 5, 6, 7. He rolls the die to create a sequence, but any roll is discarded if it sums to 8 with the most recently recorded number. Jimmy stops the moment his sequence contains at least one odd number and at least one even number. However, if he rolls a 4, the sequence is instantly wiped clean, and he must start over from scratch (the 4 is not recorded). Find the expected number of terms in Jimmy's final sequence.
15. The infinite series $S = \sum_{n=1}^{\infty} \frac{100n}{n^4 + 4}$ converges to a rational number that can be expressed as an irreducible fraction $\frac{p}{q}$. Find $p + q$.
16. In triangle ABC , $AB = 6$, $BC = 7$, and $AC = 8$. Let Γ be the circumcircle of triangle ABC . A circle ω is tangent to AB and AC and is internally tangent to Γ at a point P . The line AP intersects BC at D . Find the length of BD .
17. Let $\omega_1, \omega_2, \omega_3$ be circles all passing through a common point P with radii 3, 4, 5 and centers O_1, O_2, O_3 respectively. Let ω_1 and ω_2 intersect at A , let ω_2 and ω_3 intersect at B , and let ω_1 and ω_3 intersect at C for points A, B, C distinct from P . Let D, E, F be points on $\omega_1, \omega_2, \omega_3$ respectively such that DE passes through A , EF passes through B , and DF passes through C . If $O_1O_2 = 5$ and $O_1O_3 = 7$, the maximum possible area of DEF can be expressed as $a + b\sqrt{c}$ for positive integers a, b, c such that c is squarefree. Find $a + b + c$.
18. Let $P(x)$ be a polynomial of degree 2026^2 with distinct positive integer coefficients such that $P(0) = 1$. It is known that a polynomial $Q(x)$ of degree 2026 with distinct positive integer coefficients and $Q(0) = 2$ exists such that the product of $P(r)$ over all roots r of $Q(x)$ is equal to the product of $Q(s)$ over all roots s of $P(x)$. If m is the minimum possible value of the sum of the coefficients of $P(x)$, determine $m \bmod 1000$.

- 19.** Let $P(x)$ and $Q(x)$ be monic quadratic polynomials such that $P(Q(x))$ is minimized at $x = 1$ and $x = 2$, and $Q(P(x))$ is minimized at $x = 3$ and $x = 4$. What is the value of $P(5) + Q(5)$?
- 20.** Let $x > 2026$ be a positive integer. It is known that regardless of the value of x , the set of values $x^2, x^2 + 1, x^2 + 4, \dots, x^2 + 36$ contains a multiple of k . Determine the sum of all possible values of k .